

Course Title	Numerical Optimization		
Course Code	DS611	Credit Structure	2-0-2-3 (L-T-P-C)
Category	Core	Semester	Winter Semester (AY 2025-26)
Program	MSc(DS)		
Prerequisites	Calculus, Linear Algebra, Python		
Course Objectives/ Brief Course Description	At the end of the course, students will be able to 1. Mathematically formulate and solve different types of optimization problems 2. Define constraint functions on a problem 3. Check the feasibility and optimality of a solution 4. Apply different types of classical and modern optimization methods to solve real world engineering problems.		
Evaluation/ Grading Policy	Insem1 20% Insem2 25% Endsem 35% Lab Assignment 20%		
Course Materials/ References	1. An Introduction to Optimization, E.K.P.Chong, S.H.Zak, Wiley 2. Linear and Nonlinear Programming, Luenberger and Ye, Springer 3. Optimization for Data Analysis, Wright and Recht, Cambridge University Press 4. Practical Optimization, P.E.Gill, W.Murray, M.H.Wright, Emerald 5. Optimization in Practice with MATLAB, A Messac, Cambridge University Press		
Detailed Course Content	Basics of numerical Optimization, Mathematical formulation using Example, Unconstrained Optimization, Minimization, Maximization, Local and Global Minima, Convex sets and functions, Feasible directions, First and Second Order Necessary Conditions, Line Search Methods, Gradient method, Newton's method, Conjugate gradient method, Convergence theorems, Linear programming, Simplex method, Duality, Constrained Optimization, KKT conditions, Introduction to Genetic Algorithms		

Course Outcome:

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
x	x	x		x			x	x	x		x

Course Resources:

Lecture notes/classnotes will be made available for students after the lecture.